

Lab 1: LoRaWAN end device

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- **The Arduino project**
 - The board and the IDE
 - Arduino MKR WAN 1310
 - First example and MKRWAN library
- **LoRaWAN**
 - Topology
 - Working principles
- **The Things Network**
- **LoRaWAN End-Device registration to the network**

Arduino is an **open-source** electronics platform based on **easy-to-use hardware** and **software**.

Boards basic capabilities (much more!):

- Read inputs: digital/analog
- Set output: digital/analog

You can tell your board what to do by sending a set of **instructions** to the **microcontroller** on the board.

Arduino boards: Uno, Mega, Micro, ... MKRWAN

- Arduino IDE
- Programming language: simplified C
- The Arduino code is divided into two functions:
 - `setup()`: run only once while booting up
 - `loop()`: run cyclically
- Examples: blink
- Reset button: restarts the program



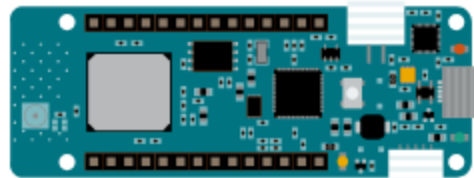
Blink (files>examples>basics>blink)

```
// the setup function runs once when you press reset
//or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(LED_BUILTIN, HIGH); // turn the LED on
  delay(1000);                      // wait for a second
  digitalWrite(LED_BUILTIN, LOW);  // turn the LED off
  delay(1000);                      // wait for a second
}
```

Arduino MKR WAN 1310

<https://docs.arduino.cc/hardware/mkr-wan-1310/>



 Buy now

MKR WAN 1310

The Arduino MKR WAN 1310 provides a practical and cost effective solution to add LoRa® connectivity to projects requiring low power. This open source board can connect to the Arduino Cloud, your own LoRa® network using the Arduino LoRa® PRO Gateway, existing LoRaWAN® infrastructure like The Things Network, or even other boards using the direct connectivity mode.

GET STARTED

DOWNLOADABLE RESOURCES

 Pinout (PDF)

 Schematics

Uploading the code

1. Load Blink example
2. Compile the code
3. Upload the code



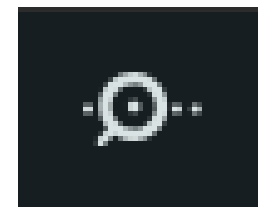
The code runs instantly, verify that a led is blinking.

Serial output:

```
println("Hello from the IoT lab\n"); //place in loop
```

Compile and upload

Open the serial monitor (top right)

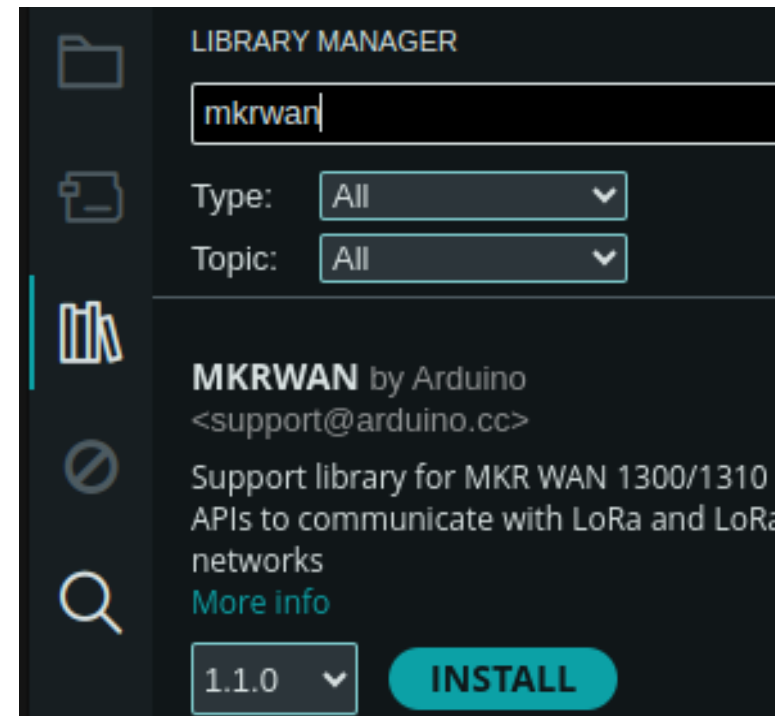


The power of Arduino: libraries

Arduino is equipped with a variety of **libraries** for whatever project you can build.

We are going to use the MKRWAN library:

- **It Implements the LoRaWAN protocol**
- **Choose MKRWAN (not MKRWAN_V2)**



"First configuration" example:

- Prints you some device info
- Tries to connect to the LoRaWAN network
 - (asks ABP or OTAA)

Let's now leave the Arduino for some minutes.

Is a protocol maintained by: **Lora Alliance**

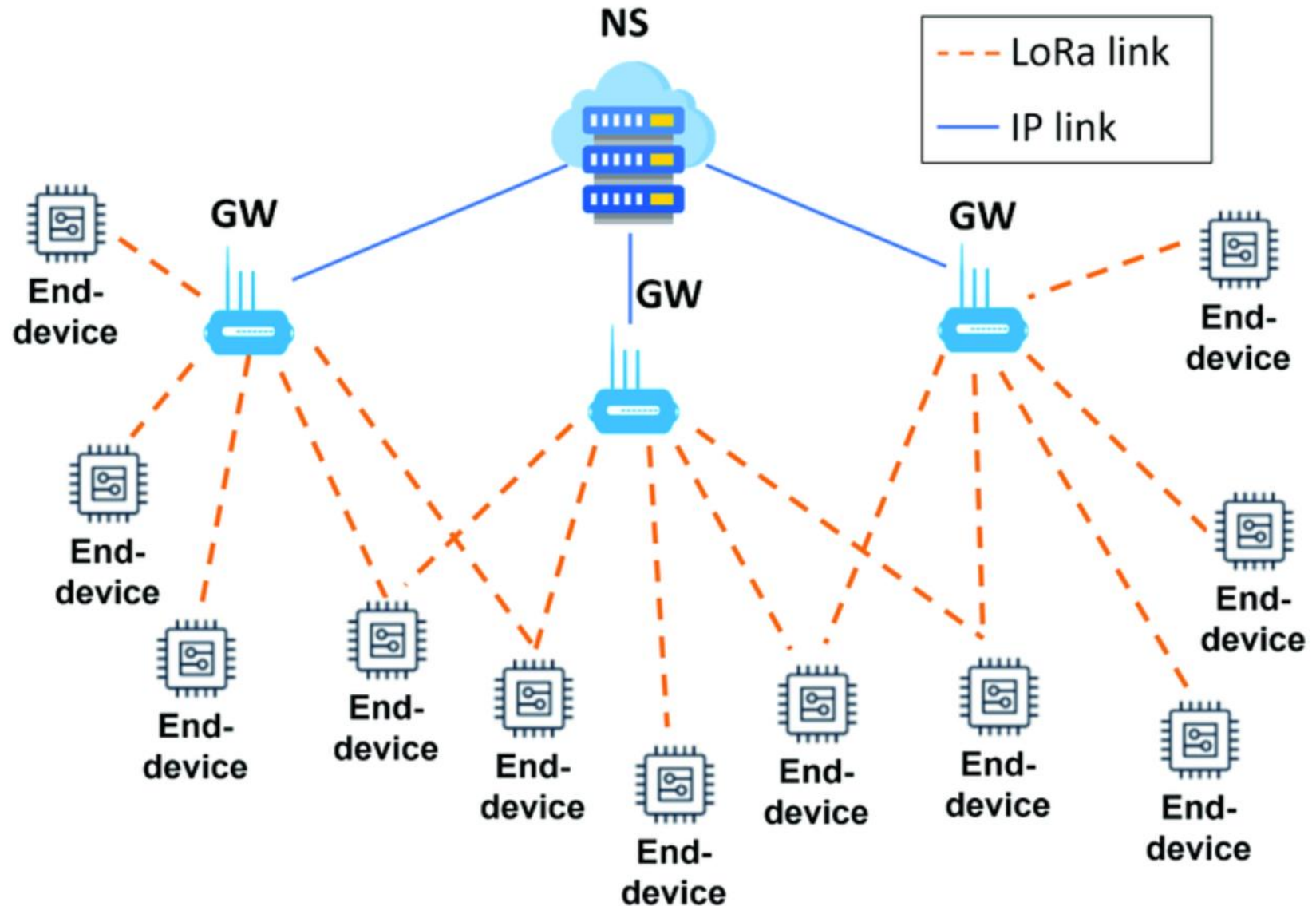
It realizes a network that enables IoT communications.

Key aspects:

- LoRa modulation
- Unlicensed frequencies
- Three components:
 - End-Devices
 - Gateways
 - Network Servers



LoRaWAN topology



The LoRa modulation allows for a tuning parameter, the Spreading Factor (SF).

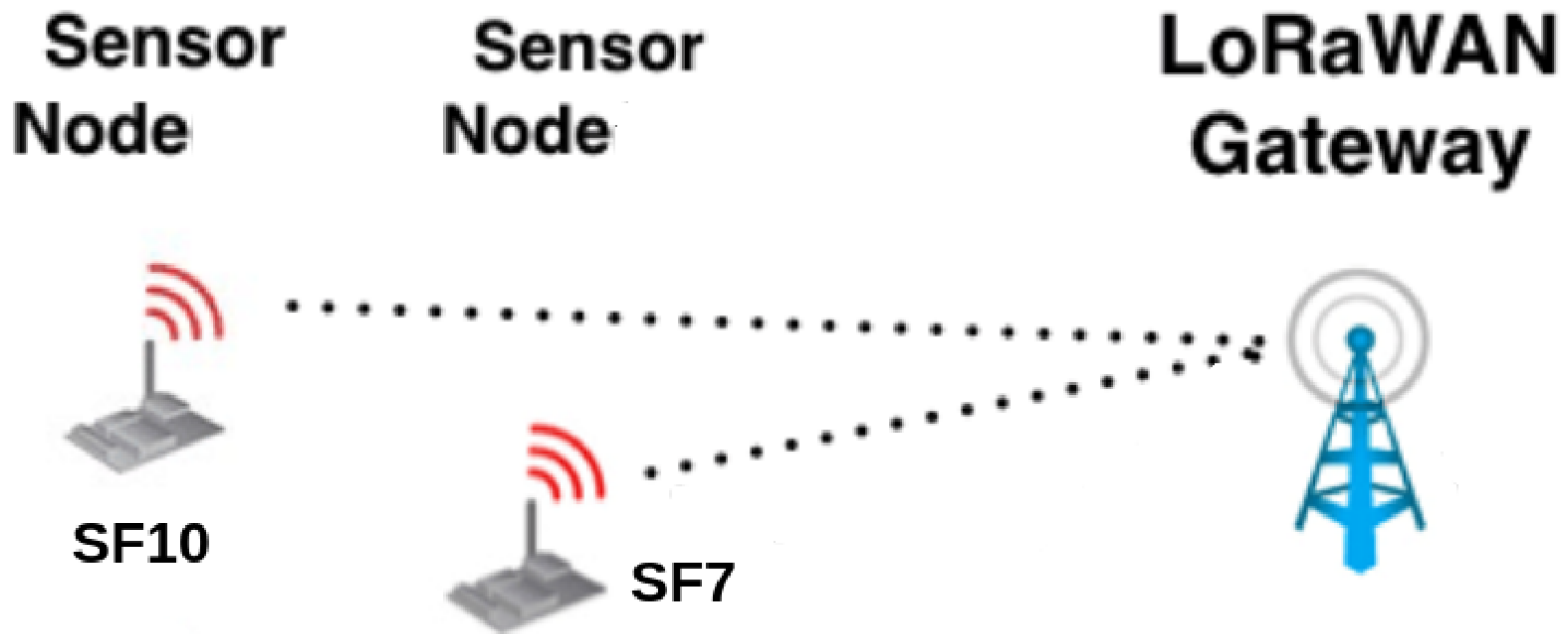
It influences **data rate, time-on-air, battery life, and distance.**

Six **SF** are available: SF7, SF8, SF12

- SF7: Faster but shorter distance
- SF12: Slower but further distance

LoRaWAN: Spreading Factor

The **SF** is **automatically tuned** to meet the **faster** communication allowed at a certain **distance**:



<https://avbentem.github.io/airtime-calculator/ttn/eu868>

(or search "lorawan airtime calculator github")

We encounter some new terms:

- "1% max duty cycle"
- "fair access policy"

And the relative number of messages per hour or day.
What do they means?

Each device, to communicate, must be **registered in the network**.

- Device EUI
 - Application EUI
 - Application key
- must be outlined.

Once registered a device can join the network.

OTAA Joining means **Over The Air Activation**.



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